

	Ballute CFD Validation Report		Type: Physical Testing	
	REPORT TITLE: Ballute CFD Validation		PROJECT	Daybreak
			SECTION	Flight Systems
ORIGINATOR:	1. Introduction Due to the high altitude flight of Project Daybreak, a novel solution is required during recovery such that the rocket is able to descend safely. Due to the lower air density at high altitudes, traditional parachutes are expected to fail as they are unable to catch enough air to provide an adequate means of slowing down. This is where the idea of a ballute is introduced. A ballute can be thought to be a tear drop shaped balloon parachute in which any air caught by the ballute is retained within its body, leading to the internal high pressure allowing it to remain expanded. A ballute is expected to be most effective at altitudes between 100,000 ft (~30km) to 3,000 ft (~1km) which is in line with the drogue usage on this rocket. Theoretical studies on the ballute have been completed, and an ideal geometry has been created. This model has one large air inlet at the bottom and 6 smaller equidistant inlets towards the middle of the ballute as shown in the following figure. 			
SECTION: Flight Systems				
DATE OF TEST:				
REPORT REVISION:				
APPROVER 1:				
DATE:				
APPROVER 2:				
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NO. PAGES:				
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FORMAT: 1. INTRODUCTION 2. OBJECTIVE 3. RESULTS 4. CONCLUSION	2. Objective The objective is to validate the theoretically calculated variables to ensure that the ballute has been accurately modelled. This would aid with the creation of physical mockups and design changes if required.			

	3. Assumptions Due to the complex geometry of the ballute and its mostly symmetrical nature, CFD has been conducted in a 2D geometry. It is assumed that the ballute retains a rigid shape. It is assumed that no airflow occurs across the inlets of the ballute. This simplification was made to reduce model complexity, and because the primary objective of the analysis was to estimate the external drag coefficient of the parachute. 4. Results Test A: <i>The simulated drag coefficient was consistent across multiple test cases, and was within 10% of the predicted value (Simulated = 0.95, Predicted = 1.05). Limitations in the setup and modelling of parachute behaviour likely contributed to the discrepancy. Ultimately, test 1 was useful in initialising the CFD model and proving computational reliability across different cases.</i> Test B: <i>The CFD model was refined (mesh quality was significantly improved and made finer). The iterative coupling method was used to converge on a final drag coefficient of 1.035 (within 2% of the predicted value). Velocity profiles indicate the ballute will perform as expected and within descent velocity requirements.</i> 5. Conclusion <i>These results strongly suggest the ballute geometry will perform as intended and should be progressed to a prototyping/testing phase. It is recommended that the CFD model is refined and used to aid in prototyping and physical testing.</i>
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Appendix A – Requirements Tested

ID	Requirement Tested

Appendix B – Test Setup, Method and Discussion

Equipment:

- ANSYS Fluent
- 2D model of a simplified ballute

Test A - Cd at Open Rocket Velocities**Description:**

Test A will be used to determine the approximate drag coefficient of the ballute at velocities predicted by OpenRocket software at specific descent altitudes. This test will primarily demonstrate the accuracy of the model (looking for consistency of results between different cases), and will be used to compare predicted Cd with Cd computed by the CFD analysis.

Setup:*Test A - Case 1*

This case represents the maximum expected velocity from initial OR simulations using proposed ballute geometry.

Altitude	90,000ft
Descent Velocity	140m/s
Air Density	0.017kg/m ³
Reference Area	1.655m ²
Reference Length	1.456m

Test A - Case 2

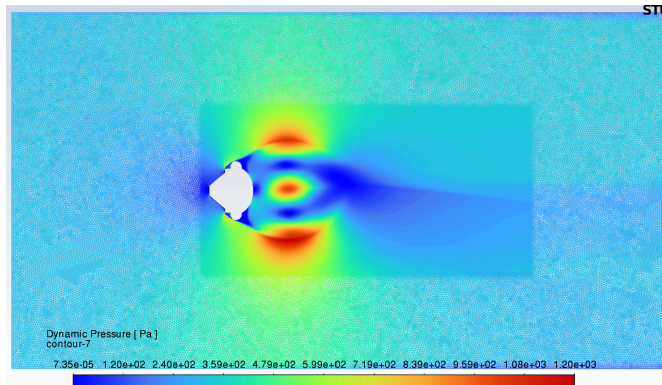
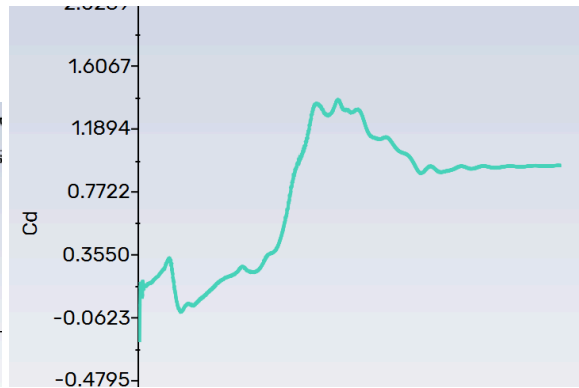
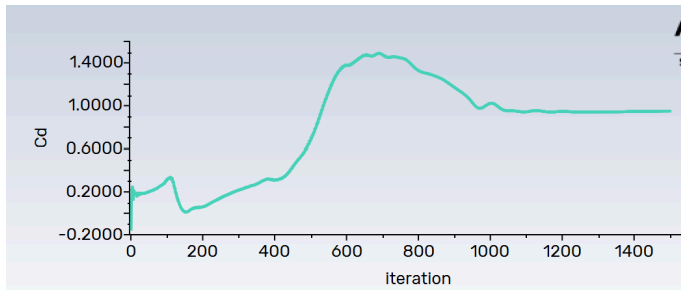
This case represents the point in the descent right before main parachute deployment. At this point, velocity will be the lowest and air density will be the highest.

Altitude	2000ft
Descent Velocity	22.86m/s
Air Density	1.155kg/m ³
Reference Area	1.655m ²
Reference Length	1.456m

Results:

The simulated drag coefficient was consistent across both test cases, and was within 10% of the value predicted by open rocket simulations (Simulated = 0.95, Predicted = 1.05). Limitations in the setup and modelling of parachute behaviour likely contributed to the discrepancy.

	Altitude	Descent Velocity	Computed Drag Coefficient
Case 1	90,000ft	140m/s	0.95
Case 2	2000ft	22.86m/s	0.95



Test B - Iterative Coupling

Description:

Since Test A indicated the reliability of the CFD model in computing consistent drag coefficients at different velocities and corresponding altitudes, Test B will use an iterative approach to converge on a more accurate drag coefficient estimate for the ballute geometry. It will use the following iterative coupling approach between mathematical analysis (using the same model used by OpenRocket), and CFD analysis.

1. Compute velocity at specified altitude (28000m) with initial drag coefficient estimate (1.05) using mathematical python model.
2. Input velocity into CFD model and compute new drag coefficient estimate.
3. Input new drag coefficient into mathematical model and compute new velocity at 28000m.
4. Input new velocity into CFD model and compute new drag coefficient estimate.
5. Repeat steps 2-4 until the approach converges on a consistent drag coefficient.

Note: It was decided to use a mathematical model in python instead of OpenRocket simulation because the OpenRocket simulation does not simulate apogee at 100,000ft. This model uses the same formulae as in OR sims, but simulates a deployment at 100,000ft. This decrease in deployment altitude also means that the calculated velocity is less than the one used in Test A (128m/s vs 140m/s). (Test A was still useful for proving the consistency of the CFD model in simulating Cd across different altitudes/velocities, and for indicating general accuracy of the predicted Cd).

CFD Setup:

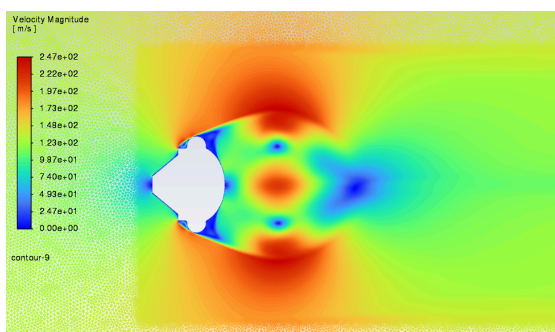
- 2D Mesh of Ballute Geometry
- Mesh was refined to improve quality (Skewness: Avg = 0.11, Max = 0.85) (Orthogonal Quality: Avg = 0.93, Min = 0.45)

Results:

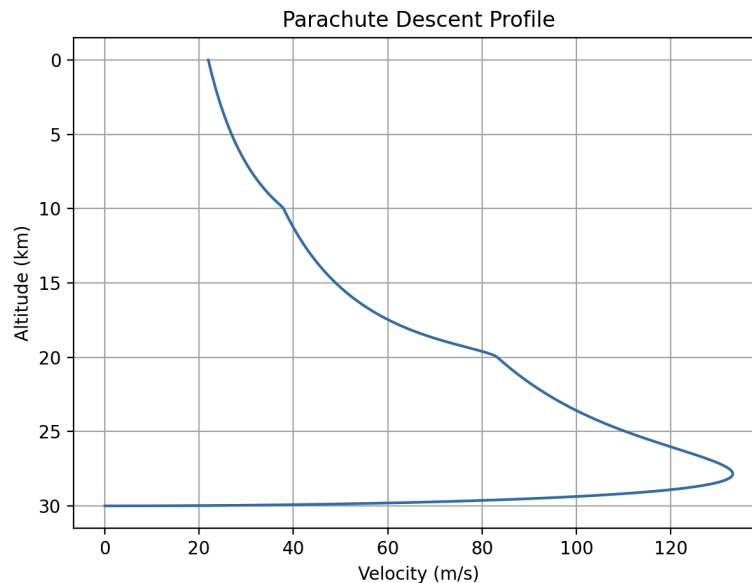
The following table outlines the iterations that were used to converge on a final coefficient of drag and velocity profile. Boxes highlighted in blue are Inputs for that case, Boxes in Green are Outputs.

	Velocity at 27000m (m/s)	Drag Coefficient
Python Script - Case 1	128.3	1.05
CFD - Case 1	128.3	1.1355
Python Script - Case 2	123.5	1.1355
CFD - Case 2	123.5	1.122
Python Script - Case 3	124.3	1.122
CFD - Case 3	124.3	1.107
Python Script - Case 4	125.1	1.107
CFD - Case 4	125.1	1.0904
Python Script - Case 5	126.0	1.0904
CFD - Case 5	126.0	1.028
Python Script - Case 6	129.5	1.028
CFD - Case 6	129.5	1.032
Python Script - Case 7	129.3	1.032
CFD - Case 7	129.3	1.035
Python Script - Case 8	129.1	1.035

See screenshots and convergence plots [here](#)

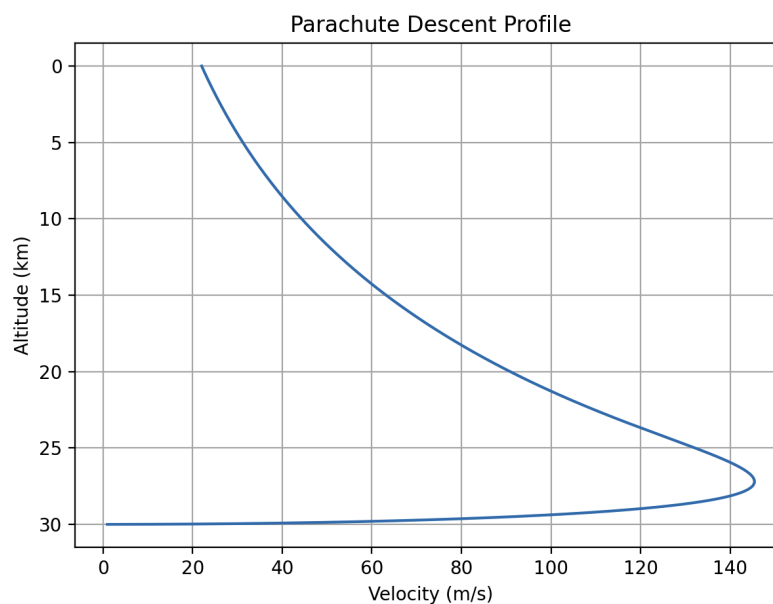


The results show a **final drag coefficient of 1.035, which is within 2% of the predicted 1.05**. This value was input into a piecewise function that mimics the openrocket simulation for descent velocity - shown below:



The above plot used a piece-wise function to mimic the OpenRocket simulation for descent velocity (OpenRocket was not used because we wanted to simulate an ideal case of apogee at 100,000ft).

Below is an alternative plot which simply uses an exponential atmospheric approximation to generate a smoother velocity profile. Both plots satisfy descent velocity requirements, hence results from this analysis suggest that the selected geometry will meet mission requirements and is ready for prototyping and testing.



Conclusion:

Overall, results suggest that the drag coefficient of the proposed geometry is consistent with the predicted value. Test A yielded drag coefficients (0.95) within 10% of the predicted value (1.05), and most importantly demonstrated the reliability of the CFD model in consistent C_d computation. Test B refined the model and utilised an iterative coupling method with the python model in order to converge on a final simulated drag coefficient of 1.035 - which is within 2% of the predicted value.

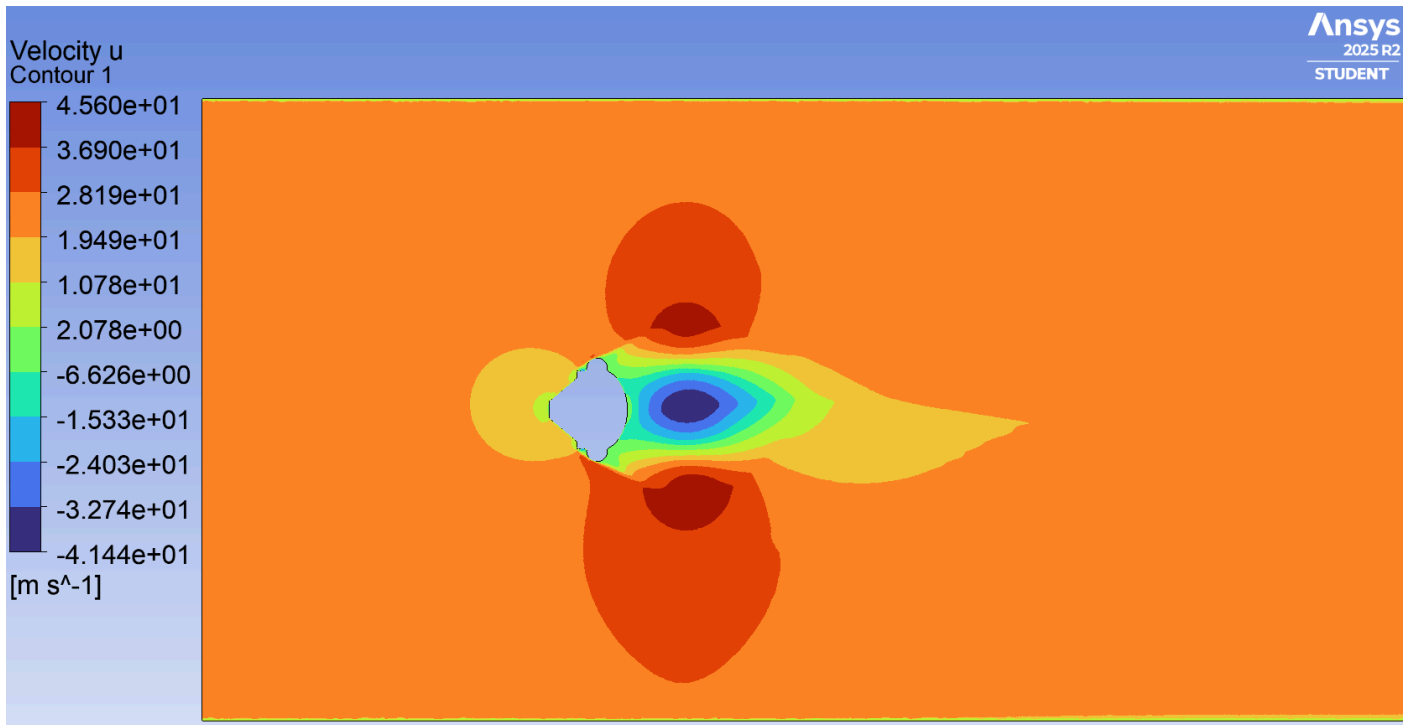
Ultimately, these results strongly support the findings of initial characterisation of the drag coefficient and suggest that the intended geometry will perform as expected and within required velocity requirements. It is recommended that the project moves to a physical testing and prototyping phase - in which the CFD model can be refined and used to inform design changes.

Further testing can be completed to include the modelling of air flow within the ballute itself by treating it as a shell object in order to ensure its shape can be maintained. However, there are significant computational limitations in approaching this problem. Other modelling can include the inflation of the ballute, particularly within the low air density regions around the rocket's apogee. This would involve the use of further software such as LS-DYNA and would require physical testing to ensure the modelling is effective.

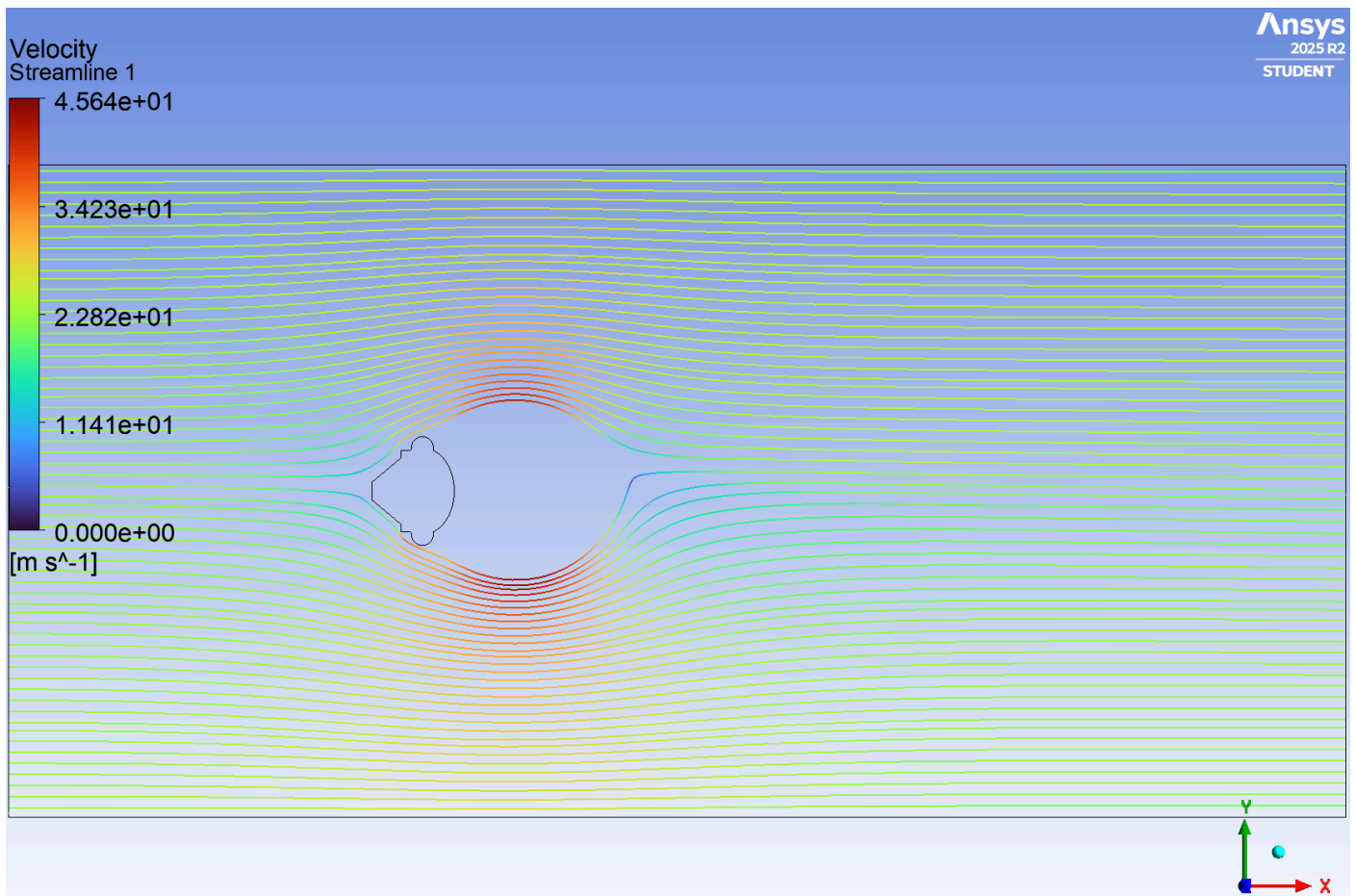
Appendix D – Files

Test A Case 2:

Contours of Velocity Magnitude



Streamlines of Velocity



Contours of Dynamic Pressure

